Causal Inference with LLMs

Day 2 — Identification & Estimation

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Day 2 — Identification & Estimation

- From Identification to Estimation
- Estimators
- Adjustment and Weighting
- LLMs and Measurement Error

Interactive App

- **URL:** Click here to open the Interactive Streamlit App
- **No Installation:** Runs entirely in the browser via Streamlit.
- **Navigation:** Use the sidebar to switch between workshop days.
- **Step-by-Step:** Each page walks through core computations.
- **Consistency:** Uses the U.S. Congressional speeches dataset (Hein Bound 111); results match local scripts.

From Identification to Estimation

Two distinct steps in causal inference:

- **Identification:**
  - Express a causal quantity in terms of observable quantities
  - Example: under ignorability, the ATE $\tau = \mathbb{E}[Y(1) - Y(0)]$ is identified from $P(Y, D, X)$
- **Estimation:**
  - Approximate the identified quantity from a finite sample
  - Specific estimators (regression, matching, IPW) are introduced in the frames that follow

Key point:

- Identification is theoretical; estimation is statistical

Identification vs Estimation (Formal Distinction)

Identification:

- A causal quantity Q is identifiable if it can be uniquely expressed as a function of the observed data distribution

Notation:

- V: collection of all observed variables (e.g., $V = (Y, D, X)$)
- $P(V)$: joint probability distribution of the observed variables

Formal idea:

- If two models imply the same observed distribution $P(V)$, they must imply the same value of Q (because the causal quantity Q is uniquely determined by the observable distribution)

Implication:

- Identification depends on assumptions, not on sample size

Note (LLM setting):

- In practice, we observe $\tilde{V} = (\tilde{Y}, D, X)$, not $V = (Y, D, X)$
- Identification results assume $P(V)$; we only have access to $P(\tilde{V})$
- This gap is the source of measurement error bias (addressed in Part 2)

Estimand, Estimator, Estimate

Estimand:

- The target causal quantity in the population (e.g., $\tau = \mathbb{E}[Y(1) - Y(0)]$)

Estimator:

- A rule that maps observed data $\{(Y_i, D_i, X_i)\}_{i=1}^n$ to a numerical value

Estimate:

- The realized value of the estimator for a given sample

Distinction:

- Estimand: population quantity
- Estimator: mathematical rule
- Estimate: number computed from data

Notation Guide

Superscripts and decorations:

Symbol	Meaning
τ	True (population) causal quantity — the estimand
$\hat{\tau}$	Estimated value computed from data — the estimate
$\tilde{Y}_i$	LLM-measured (noisy) version of outcome Y_i
$\hat{m}_d(x)$	Estimated conditional expectation function
$\hat{e}(X_i)$	Estimated propensity score

Key distinction:

- $\hat{\cdot}$ (hat): estimated from data — a statistical approximation
- $\tilde{\cdot}$ (tilde): measured by LLM — a noisy proxy for the true variable
- No decoration: true population quantity or true latent variable

Data Structure for Estimation

Observed data:

$$\{(Y_i, D_i, X_i)\}_{i=1}^n$$

Notation:

- $i = 1, \ldots, n$: index of units
- n: sample size (number of observed units)
- (Y_i, D_i, X_i): observed outcome, treatment, covariates for unit i

Goal:

- Estimate the ATE τ from observed data to obtain an estimator $\hat{\tau}$

Sampling Variability and Finite Samples

Observation:

- The estimator $\hat{\tau}$ depends on the particular sample $\{(Y_i, D_i, X_i)\}_{i=1}^n$

Key distinction:

- **Systematic bias:** due to confounding, selection, or measurement error
- **Random variability:** due to finite sample size

Implication:

- Even if identification holds (no systematic bias), estimates may differ from τ
- The discrepancy decreases as n increases

Interpretation:

- Causal inference requires controlling both:
  - bias (via assumptions)
  - variability (via statistical estimation)

Unbiasedness of Estimators

Definition:

- An estimator $\hat{\tau}$ is **unbiased** if

$$\mathbb{E}[\hat{\tau}] = \tau$$

Interpretation:

- On average over repeated samples, the estimator recovers the true ATE

Important distinction:

- Unbiasedness does not imply accuracy in a given sample
- Variability may still be large when n is small

Implication:

- Both unbiasedness and low variability are required for reliable estimation

Difference-in-Means Estimator

$$\hat{\tau}_{\text{diff}} = \bar{Y}_1 - \bar{Y}_0$$

Notation:

- $\bar{Y}_1 = \frac{1}{n_1} \sum_{i:D_i=1} Y_i$: average outcome among treated units
- $\bar{Y}_0 = \frac{1}{n_0} \sum_{i:D_i=0} Y_i$: average outcome among control units
- $n_1 = \sum_{i=1}^{n} D_i, \quad n_0 = \sum_{i=1}^{n} (1 - D_i)$

Interpretation:

- Compares average outcomes between treated and control groups

Key point:

- Unbiased for the ATE τ under random assignment of treatment

Assumes:

1. Random assignment of treatment
2. SUTVA (no interference between units, single version of treatment)
3. Perfect measurement: Y_i is observed directly (without LLM-induced error)

- Note: assumption 3 fails when Y_i is measured by an LLM

Sampling Variance of the Difference-in-Means

$$\mathrm{Var}(\hat{\tau}_{\text{diff}}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}$$

Notation:

- S_1^2: variance of potential outcomes $Y(1)$ in the population
- S_0^2: variance of potential outcomes $Y(0)$ in the population
- S_τ^2: variance of individual treatment effects $Y(1) - Y(0)$

Interpretation:

- Variability arises from heterogeneity in outcomes and treatment effects

Key point:

- Sampling variability persists even under perfect randomization

Why Adjustment is Needed

Observation:

- The difference-in-means estimator compares groups defined by $D_i = 1$ (under treatment) and $D_i = 0$ (without treatment)

Problem:

- If treatment is not randomly assigned, these groups may differ systematically in X_i

Consequence:

- Differences in outcomes may reflect both:
  - causal effects of D_i
  - pre-existing differences in X_i

Goal:

- Adjust for X_i to isolate the causal effect of D_i on Y_i

Regression Adjustment

Why regression is needed:

- Treatment groups ($D_i = 1$ vs $D_i = 0$) may differ in covariates X_i
- Direct comparison of outcomes mixes causal effects with differences due to X_i

Idea:

- Adjust for X_i by modeling the outcome conditional on treatment and covariates

Specification:

$$Y_i = \alpha + \tau D_i + \beta^\top X_i + \varepsilon_i$$

Notation:

- α: intercept (baseline outcome)
- τ: coefficient measuring the effect of treatment
- β: vector of coefficients for covariates X_i
- ε_i: error term capturing unexplained variation

Interpretation:

- τ measures the average difference in outcomes between treated and control units, holding X_i fixed

Conditions for Valid Regression Adjustment

Key requirement:

- The regression model must correctly adjust for confounding variables X_i

Interpretation:

- After conditioning on X_i, the remaining variation in D_i is as good as random

Connection to ignorability:

- Validity requires:

$$(Y(1), Y(0)) \perp D \mid X$$

Implication:

- If relevant confounders are omitted from X_i, the estimate of τ will be biased

Propensity Score as a Balancing Score

$$e(X_i) = P(D_i = 1 \mid X_i)$$

Definition:

- The propensity score is the probability that unit i receives treatment given its covariates X_i

Balancing property:

- Conditional on $e(X_i)$, the distribution of X_i is the same in treated and control groups

Interpretation:

- Units with the same propensity score are comparable, even if their covariates differ

Implication:

- Matching or weighting on $e(X_i)$ removes systematic differences in X_i

Conditional Expectation as a Regression Function

$$m_d(x) := \mathbb{E}[Y \mid D = d, X = x]$$

Interpretation:

- $m_d(x)$ is the average outcome among units with:
  - treatment status $D = d$
  - covariates equal to $X = x$

Key properties:

- $m_1(x) - m_0(x)$: **conditional average treatment effect (CATE)** at $X = x$
- Averaging over X: $\tau = \mathbb{E}_X[m_1(X) - m_0(X)]$

Example:

- W_i = speech text; $Y_i = f(W_i)$ sentiment score derived from speech text W_i; D_i = party (R=1, D=0);
- $m_1(x)$: expected sentiment for a Republican with covariates $X = x$
- $m_0(x)$: expected sentiment for a Democrat with covariates $X = x$

Measured version (LLM):

- $\tilde{m}_d(x) = \mathbb{E}[\tilde{Y}_i \mid D_i = d, X_i = x]$
- In practice we estimate $\tilde{m}_d(x)$, not $m_d(x)$

Role:

- Central object in regression-based estimation

Plug-in Estimator of the ATE

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^{n} (\hat{m}_1(X_i) - \hat{m}_0(X_i))$$

Notation:

- $\hat{m}_d(x)$: estimate of $m_d(x)$ obtained from data (e.g., via regression)
- X_i: observed covariates of unit i
- $\hat{m}_1(X_i)$: predicted outcome for unit i under treatment
- $\hat{m}_0(X_i)$: predicted outcome for unit i under control
- $\hat{m}_1(X_i) - \hat{m}_0(X_i)$: estimated conditional treatment effect at X_i

Overall interpretation:

- Evaluate the estimated conditional treatment effect at each observed X_i
- Average across units to obtain an estimate of τ

Plug-in Estimator and Regression

Observation:

- The plug-in estimator requires estimating the functions

$$m_d(x) = \mathbb{E}[Y \mid D = d, X = x], \quad d \in \{0, 1\}$$

Implementation:

- In practice, $m_d(x)$ is estimated using regression methods
- Examples:
  - linear regression
  - nonparametric or machine learning models

Connection:

- Regression adjustment provides estimates $\hat{m}_1(x)$ and $\hat{m}_0(x)$
- These are then used in the plug-in formula for $\hat{\tau}$

Key point:

- Regression is not an alternative to the plug-in estimator, but a way to implement it

Matching

Idea:

- Compare treated units ($D_i = 1$) with control units ($D_i = 0$) that have similar covariates X

Procedure:

- For each treated unit, find one or more control units with similar X
- Compare their observed outcomes

Goal:

- Approximate a randomized experiment by balancing covariates between treated and control groups

Estimator:

$$\hat{\tau}_{\text{match}} = \frac{1}{n_1} \sum_{i:\, D_i=1} \left(Y_i - Y_{j(i)} \right)$$

- $j(i)$: index of the matched control unit for treated unit i
- $Y_{j(i)}$: observed outcome of the matched control unit

Matching as Conditional Comparison

Objective:

- Approximate the conditional expectation

$$\mathbb{E}[Y \mid D = d, X = x]$$

  by comparing units with similar covariates X

Mechanism:

- For a treated unit with covariates X_i, select control units with covariates X_j close to X_i
- Use their outcomes to approximate the counterfactual outcome

Interpretation:

- Matching constructs local comparisons at fixed X
- These comparisons approximate the functions $m_1(x)$ and $m_0(x)$

Connection:

- Matching provides a nonparametric way to estimate the components of the plug-in estimator

Estimating the Propensity Score

In practice:

- The true propensity score $e(X_i) = P(D_i = 1 \mid X_i)$ is unknown
- It must be estimated from data

Standard approach — logistic regression:

$$\log \frac{\hat{e}(X_i)}{1 - \hat{e}(X_i)} = \hat{\alpha} + \hat{\beta}^\top X_i$$

Procedure:

- Regress D_i on X_i using logistic regression
- Use predicted probabilities $\hat{e}(X_i)$ in place of $e(X_i)$

Note:

- More flexible models (random forests, neural nets) can also be used
- Misspecification of $e(X_i)$ leads to biased IPW estimates

Inverse Probability Weighting (IPW)

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{D_i Y_i}{e(X_i)} - \frac{(1-D_i)Y_i}{1-e(X_i)}\right)$$

Idea:

- Give more weight to units that are underrepresented given their covariates

Interpretation:

- Reweight observations to create a pseudo-population
- In this pseudo-population, treatment is independent of covariates
- Thus, comparison between treated and control mimics a randomized experiment

Note:

- Requires overlap: $0 < e(X_i) < 1$ (otherwise weights explode)

Why Inverse Probability Weights?

Problem:

- Some covariate profiles X_i are more likely to receive treatment than others
- This creates imbalance between treated and control groups

Idea:

- Reweight observations to compensate for unequal probabilities of treatment

Mechanism:

- Units with low probability of their observed treatment receive higher weight
- Units with high probability receive lower weight

Interpretation:

- Creates a synthetic sample in which treatment assignment is independent of X_i

IPW as Reweighting the Data

Population identities:

$$\mathbb{E}[Y(1)] = \mathbb{E}\left[\frac{D\,Y}{e(X)}\right], \qquad \mathbb{E}[Y(0)] = \mathbb{E}\left[\frac{(1-D)\,Y}{1-e(X)}\right]$$

Implication:

- The ATE can be written as

$$\tau = \mathbb{E}\left[\frac{D\,Y}{e(X)} - \frac{(1-D)\,Y}{1-e(X)}\right]$$

Sample analogue:

- Replace expectations by sample averages:

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{D_i Y_i}{e(X_i)} - \frac{(1-D_i)Y_i}{1-e(X_i)}\right)$$

Key point:

- The IPW estimator is the empirical version of these weighted expectations

Augmented IPW (Doubly Robust) Estimator

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\underbrace{\hat{m}_1(X_i) - \hat{m}_0(X_i)}_{\text{Outcome Model}} + \underbrace{\frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-D_i)(Y_i - \hat{m}_0(X_i))}{1-\hat{e}(X_i)}}_{\text{Weighted Residual}}\right]$$

Key property (double robustness):

- Consistent if *either* $\hat{m}_d(x)$ *or* $\hat{e}(X_i)$ is correctly specified
- Does not require both models to be correct simultaneously

Interpretation:

- Starts from the plug-in (G-formula) estimate
- Adds IPW correction terms to remove bias from outcome model misspecification

Why it matters:

- In practice, neither model is perfectly specified
- AIPW provides insurance against one of the two being wrong

From Estimation to Measurement

So far:

- We assumed that variables (Y_i, D_i, X_i) are directly observed
- Estimation methods (regression, matching, IPW) rely on these variables

In many applications:

- Variables are not directly observed
- Instead, they must be extracted from raw data (e.g., text, images)

Implication:

- The quality of causal estimation depends on how these variables are measured

Transition:

- We now introduce LLMs as tools for constructing (Y_i, D_i, X_i) from raw data

Measurement with LLMs

$$\tilde{Y}_i = f_{\text{LLM}}(W_i)$$

Notation:

- W_i: raw data (e.g., text) for unit i
- $\tilde{Y}_i$: LLM-based measurement of the outcome
- Y_i: true (latent) outcome

Key issue:

- $\tilde{Y}_i \neq Y_i$ (measurement error)
- Errors may be systematic and depend on W_i or D_i

Example (congressional speeches):

- W_i = text of speech i
- Y_i = true sentiment of the speech (latent)
- $\tilde{Y}_i$ = sentiment score assigned by the LLM
- The LLM may score speeches differently depending on party affiliation D_i, introducing systematic error correlated with treatment

Latent vs Measured Variables

Distinction:

- Y_i, D_i, X_i: target variables of interest
- $\tilde{Y}_i, \tilde{D}_i, \tilde{X}_i$: variables constructed by the LLM

In our application:

- Only Y_i is LLM-measured ($\tilde{Y}_i$); D_i and X_i are observed directly

Interpretation:

- Causal inference is defined in terms of (Y_i, D_i, X_i)
- In practice, we observe $(\tilde{Y}_i, \tilde{D}_i, \tilde{X}_i)$

Implication:

- Estimation is performed on measured variables, not on the true ones
- Discrepancies between the two induce measurement error

Measurement Error and Bias

$$\tilde{Y}_i = Y_i + \varepsilon_i$$

Classical case (MCAR — Missing Completely At Random):

- $\varepsilon_i \perp (D_i, X_i)$
- Measurement error is mean-independent of treatment and covariates
- Inflates variance but does *not* bias the ATE

LLM case (MNAR — Missing Not At Random):

- ε_i may depend on treatment D_i or covariates X_i
- $\Rightarrow$ non-classical (systematic) measurement error
- LLMs are often MNAR because:
  - Prompt sensitivity: the LLM response may vary with the content of W_i, which is correlated with D_i or X_i
  - Training bias: LLMs may have absorbed systematic biases that correlate ε_i with D_i or X_i

Consequence:

- $\mathbb{E}[\varepsilon_i \mid D_i] \neq 0 \;\Rightarrow$ biased estimates of treatment effects

Implications for Estimation

- Regression estimates may be biased
- Propensity scores $e(X_i)$ may be misestimated
- Ignorability may fail after measurement

Why:

- Measurement error can introduce dependence between D_i and potential outcomes

Key message:

- Measurement is part of the causal model, not a preprocessing step
- LLM outputs must be modeled, not taken as ground truth

Preview: Correcting Measurement Error

The problem:

- We observe $\tilde{Y}_i$ (LLM output) instead of Y_i (true outcome)
- Under MNAR, naive estimation using $\tilde{Y}_i$ yields biased causal estimates

Correction strategy:

- Use a small labeled dataset where both Y_i and $\tilde{Y}_i$ are observed
- Use labeled data to calibrate LLM outputs
- Combine statistical correction with causal estimation
- Recover valid causal effects under assumptions

Next step (Day 3):

- Design-Based Supervised Learning (DSL)

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Questions and Discussion

Thank you!

Questions?

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